DEEPIKA PODICHETTY

AI/ML Engineer | Generative AI Engineer | Associate Forward Deployed Engineer
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PROFESSIONAL SUMMARY

AI/ML and Generative AI Engineer with 4+ years of software engineering experience, building and shipping enterprise AI products at Solix Technologies across data intelligence, natural language querying, agentic platforms, voice AI, and federated data access. Full-stack contributor to LLM applications with strong expertise in RAG, knowledge graphs, prompt engineering, multi-agent workflow design, and LLM evaluation. Growing client-facing experience delivering technical demos for enterprise accounts including McDonald's. Prior foundation in enterprise .NET engineering at TCS.

TECHNICAL SKILLS

Generative AI & LLMs: GPT-4o, Claude 3.5, Llama 3, Mistral, Gemini, Azure OpenAI, Structured Outputs, Function Calling, Prompt Engineering, Guardrails, Fine-Tuning (LoRA/QLoRA), LLM Evaluation

Agentic AI: LangChain, LangGraph, CrewAI, AutoGen, Multi-Agent Systems, Sequential/Parallel/Hybrid Workflows, Tool Calling, ReAct, LangSmith

RAG & Retrieval: RAG, Agentic RAG, Knowledge Graphs, Hybrid Search, Re-ranking, LlamaIndex; ChromaDB, Pinecone, FAISS, Azure AI Search, Weaviate, pgvector, Qdrant

ML, NLP & Voice: K-Means, XGBoost, NER, Text-to-SQL, OCR, PyTorch, TensorFlow, Scikit-learn, HuggingFace; OpenAI Whisper, Piper TTS, LIDA, Plotly 3D

Backend & Cloud: Python, SQL, C#, JavaScript; FastAPI, REST, Celery, SSE/WebSockets; Azure, AWS (SageMaker, Bedrock), GCP (Vertex AI), Docker, Kubernetes, GitHub Actions, MLflow, vLLM, Ollama

Evaluation & Governance: LangSmith, Langfuse, RAGAS, DeepEval, Grafana, Prometheus, Responsible AI, Red Teaming, RBAC, Audit Logging

Familiar & Exploring: MCP, A2A Protocol, GraphRAG, Apache Kafka, OpenTelemetry, Terraform

PROFESSIONAL EXPERIENCE

Associate Forward Deployed Engineer | AI/ML & Generative AI Engineer, Solix Technologies Inc. *Oct 2024 – Present*

- Building a suite of 7 enterprise AI products (DataSense, DataAsk, AI Agent Studio, Jessica, Semantic Layer, AI Kernel, AI Data Exchange) as a full-stack contributor.
- Contributed to RAG, knowledge graph, and vector retrieval pipelines in DataSense, combining LLMs with K-Means and XGBoost to classify data sensitivity, categories, and retention policies.
- Built and refined Text-to-SQL in DataAsk (LangChain, structured outputs, LangSmith); integrated LIDA and Plotly 3D for natural language to chart output.
- Proposed and implemented parallel and hybrid multi-agent workflow patterns and a dual-mode (user-designed or AI-generated) workflow design in AI Agent Studio.
- Built Jessica, the Solix AI voice assistant, using OpenAI Whisper and Piper TTS with an LLM backend.
- Contributing to pharma/healthcare AI user stories, including InChI key and SMILES processing for scientific datasets.
- Conduct client technical demos and walkthroughs for enterprise accounts including McDonald's, leading one engagement and supporting two more in a mentored delivery model.

DevOps Engineer, Origin Hubs Inc. *Jul 2024 – Oct 2024*

- Built CI/CD pipelines (Jenkins, GitHub Actions), managed Docker/Kubernetes workloads, and set up Grafana and Prometheus monitoring.

.NET Developer, Tata Consultancy Services (Protective Life Insurance) *Oct 2019 – Aug 2022*

- Built enterprise insurance applications (ASP.NET MVC, C#, REST APIs) with multi-tier architectures; modernized document processing using OCR and rule-based NLP.
- Optimized SQL Server performance (stored procedures, index tuning) and delivered feature releases in Agile/Scrum sprints.

KEY PROJECTS — SOLIX TECHNOLOGIES

- **DataSense:** Data intelligence and classification platform (RAG, knowledge graphs, K-Means, XGBoost, ChromaDB, Azure AI Search); also the knowledge base powering DataAsk.
- **DataAsk:** Natural language data querying (Text-to-SQL, LangChain, LangSmith, LIDA, Plotly 3D, PostgreSQL, Azure OpenAI) returning tables, charts, and 3D visuals; supporting integration into Solix's Common Data Platform.
- **AI Agent Studio:** No-code platform to create agents and workflows via natural language (LangGraph, CrewAI, AutoGen, FastAPI, React, Docker, Azure).
- **Semantic Layer, AI Kernel, AI Data Exchange:** Federated data query platform with built-in governance; AI asset review marketplace; and governed enterprise data marketplace.

EDUCATION

MS in Computer Science, Western Illinois University (GPA 3.8/4.0) *May 2024*

BTech in Computer Science, Jawaharlal Nehru Technological University (GPA 3.4/4.0) *May 2019*

CERTIFICATIONS & AWARDS

- Award for Excellence (x2), Innovation Pride Award, SEAL 1.8 Hackathon Winning Team, Xplore 2 One-Time Incentive (Tata Consultancy Services)
- Completed: DeepLearning.AI LangChain for LLM Application Development; Building Systems with ChatGPT API; LLM Evaluation & Monitoring
- Pursuing: AWS Certified ML Specialty; Azure AI Engineer (AI-102); DeepLearning.AI LLMOps Specialization